

Wearable robot companion

Loneliness can hurt the human mind but what if you have a robot companion always there for you? Well, researchers at Yamagata University, Japan, are developing a robot called MH-2, which is a wearable miniature humanoid that sits on your shoulder and can be remotely inhabited by your relatives or friends from anywhere in the world.



Wearable MH-2 robot by Yamagata University that sits on your shoulder and can be remotely inhabited by your relatives or friends from anywhere in the world.

The wearable telepresence robot acts as an avatar for a remote operator. With two arms having seven degrees of freedom (DOFs), a 3-DOF head and 2-DOF body, plus one additional DOF for realistic breathing, the MH-2 is designed to mimic human actions as accurately and realistically as possible, according to an IEEE report.

While you wear the MH-2, your remote sitting friend puts on a 360-degree immersive 3D display and then stands in front of a motion capturing environment like a Kinect. By doing so, he is able to see whatever the MH-2 sees. Meanwhile, the robot on your shoulder duplicates the speech and gestures of your friend right there for

you to interact with directly.

Computers to read your facial expressions

MIT researchers are working on a new technology that could allow computers to classify human facial expressions of frustration and genuine delight. This technology may prove to be of great help for people with autism. It would help them learn to figure out and understand expressions accurately. The



cations. It is an important reminder that not all smiles are positive. There has been a tendency to 'read' enjoyment whenever smiles are found. For human-computer interaction, among other fields and applications, a more nuanced view is needed."

New glasses help visually impaired detect obstacles

UC3M scientists have developed a simple device that would help visually impaired people to perceive the full extent of their surroundings. A prototype system, based around a head-mounted display and a pair of small cameras, has been designed.

A small computer is attached with the head-mounted display and a pair of small cameras, which processes all the images it receives. Through the designed algorithm, the system actuates the distances and outlines the objects and then communicates this information to the user in real time. It uses two micro screens and highlights the silhouette of the objects in the picture and varies the colours corresponding to their distances.

"It detects objects and people who move within the visual field that a person with no visual pathologies would have. Very often the patient does not detect them due to problems of contrast. The information regarding depth is what is most missed by patients who use this type of technical aid," Prof. Vergaz added.

The scientists are trying to make the computer portion smaller and more portable, making it less burdensome for the users.

Solar cells on paper

Researchers at the Chemnitz University of Technology (CUT), Germany, have developed solar cells that are printed on paper. This technology, known as 3PV, uses conventional printing methods and standard type

researchers plan to break the expressions into low-level features in order to help computers assess mood.

Jeffrey Cohn, a professor of psychology at the University of Pittsburgh, says this work "breaks new ground with its focus on frustration—a fundamental human experience. While pain researchers have identified smiling in the context of expressions of pain, the MIT group may be the first to implicate smiles in expressions of negative emotion."

Cohn adds, "This is a very exciting work in computational behavioural science that integrates psychology, computer vision, speech processing and machine learning to generate new knowledge with clinical impli-